

Total No. of Questions : 6]

SEAT No. :

P106

[Total No. of Pages : 2

Oct.-16/BE/Insem.- 164

B.E. (Information Technology) (Elective - I (d))

PARALLEL ALGORITHM & DESIGN

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Figure to the right indicate full marks.*

- Q1)** a) What are parallel Algorithms? [2]
b) What is an Algorithm design technique? [2]
c) Discuss PRAM model of parallel computation. [6]

OR

- Q2)** a) Differentiate between the hypercube & cube connected cycle parallel computational model. [6]
b) Explain sequential Algorithms with an example. [4]

- Q3)** a) Explain Divide & Conquer Algorithm. State a sorting algorithm using same algorithm. [8]
b) What is cost of PA? Explain. [2]

OR

- Q4)** a) Write algorithm for performing summation $S = A(1) + A(2) + \dots + A(n)$ for shared memory model. [6]
b) Write a short note on systolic Algorithms. [4]

P.T.O.

- Q5)** a) Explain Parallel Merging Networks. [6]
b) Discuss the Quick Sort Algorithm in detail. [4]

OR

- Q6)** a) How to implement a parallel sorting algorithm on EREW model? [6]
b) Give a suitable example & explain the Breadth first search. [4]

